

Math 290

Practice Quiz 2 — Version G

(extra practice, Sections 6–11)

1. Short response.

- (a) Evaluate the following proposed theorem and proof.

Theorem: If $x \in \mathbb{Q}$ and $y \in \mathbb{R} - \mathbb{Q}$, then $x + y \in \mathbb{R} - \mathbb{Q}$.

Proof. Let $x = \frac{1}{2} \in \mathbb{Q}$ and $y = \sqrt{2} \in \mathbb{R} - \mathbb{Q}$. Then $x + y = \frac{1}{2} + \sqrt{2}$, which is irrational. Hence the theorem holds. $\square$

- A. The theorem is true and the proof is correct.
 - B. The theorem is false.
 - C. The theorem is true, but the proof is incorrect because checking one example does not prove a universal statement.
 - D. The theorem is true, but the proof is incorrect because it should have been done contrapositively.
 - E. The theorem is true, but the proof is incorrect because $\frac{1}{2} + \sqrt{2}$ is rational.
- (b) True or false: $x = 0$ is a counterexample to the statement “For all $x \in \mathbb{R}$, $x > \frac{1}{2}x^2$.”
- (c) True or false: Let $a, b, c \in \mathbb{Z}$. The converse of the implication “If $a \nmid bc$, then $a \nmid b$ and $a \nmid c$.” is true.
2. Let $a, b, c, d \in \mathbb{Z}$. Prove that if $a \mid c$ and $b \mid d$, then $ab \mid cd$. Then state the contrapositive of this implication.
3. Prove that there exist irrational numbers a and b such that a^b is irrational. (You may use that $\sqrt{2}$ is irrational, that a nonzero rational number times an irrational number is irrational, and that a rational number plus an irrational number is irrational. Hint: $\sqrt{2}^{\sqrt{2}} \cdot \sqrt{2}^{1-\sqrt{2}} = \sqrt{2}$.)
4. For each $n \in \mathbb{N}$, define $S_n = \{m \in \mathbb{Z} : m \leq n\}$. (Recall that $\mathbb{N} = \{1, 2, 3, \dots\}$.)
- (a) Prove that $\bigcup_{n \in \mathbb{N}} S_n = \mathbb{Z}$.
 - (b) Determine $\bigcap_{n \in \mathbb{N}} S_n$, and justify your answer.